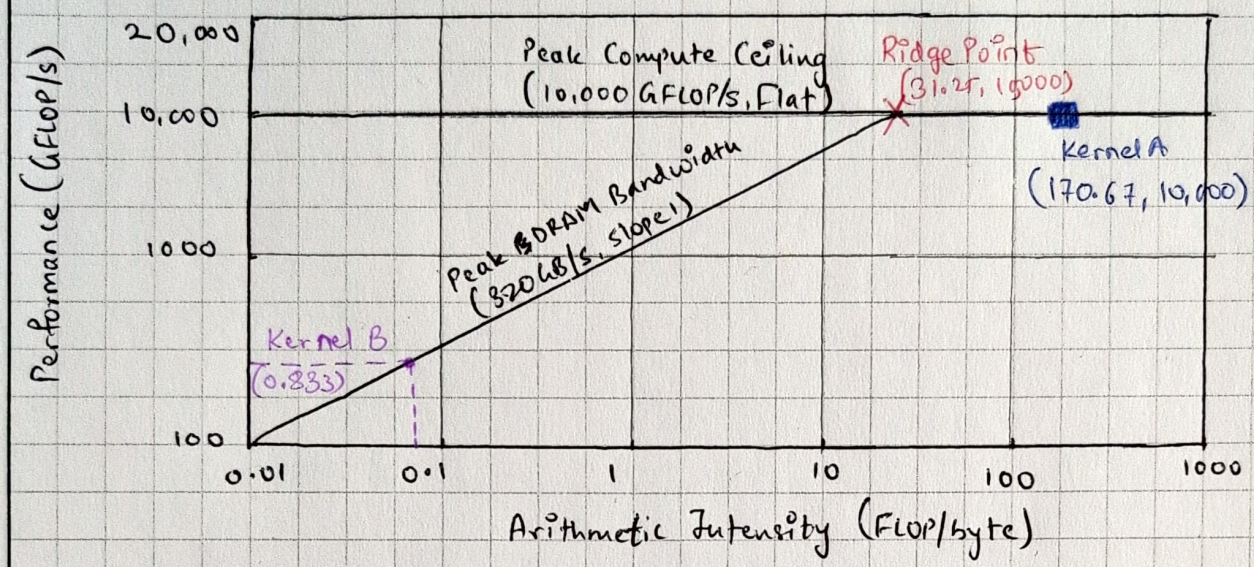
AMOGH THAKUR

HARDWARE FOR AI

ECE 510 – CMAN2

Given- Peak Compute = 10 TFLOPS (FP32) Peak DRAM Bandwidth = 320GB/s

Ridge point calculation. $= \frac{10{,}000}{320GB/s} \cong 31.25$ FLOP/byte.



Kernel A. (Dense GEMM 1024×1024).

FLOPS – $2 \times N^3 = 2 \times 1024^3$
$= \underline{2147483648 \text{ FLOPS}}$

Bytes – $3 \times N^2 \times 4 = 3 \times 1024^2 \times 4$
$= \underline{12{,}582{,}912 \text{ Bytes.}}$

Arithmetic Intensity $= \frac{\text{FLOPs}}{\text{Bytes}}$
$= \frac{2147483648}{12582912}$

Arithmetic Intensity $= \underline{170.67 \text{ FLOPS/Byte.}}$

Bound – Compute Bound (AI > 31.25)
↑ Ridgepoint

Attainable Performance Ceiling – 10,000GFLOP.

Architectural Recommendation – Increase Peak Compute Performance (TFLOPs) through architectural scaling / faster clock.

Kernel B (Vector Addition length-4194304)

FLOPS – Length $= \underline{4{,}194{,}304 \text{ FLOPS}}$

Bytes – 3×Length×4 $= 12 \times 4{,}194{,}304$
$= \underline{50{,}331{,}648 \text{ Bytes.}}$

Arithmetic Intensity $= \cancel{\frac{\text{FLOPS}}{\text{Bytes}}}\ \frac{\text{FLOPS}}{\text{Bytes}}$

But here as we consider only 1 addition per index (L = 1)

LOAD + LOAD + STORE = 4 + 4 + 4 = 12

$\therefore$ AI $= \frac{L}{\text{Bytes}} = \frac{1}{12} \cong \underline{0.0833 \text{ F/B}}$

Bound – Memory Bound (AI < 31.25)
↑ Ridgepoint.

Attainable Performance Ceiling = BW × AI
$= \underline{320 \text{ GB/s} \times 0.0833}$

Architectural Recommendation – Increase Peak DRAM BW (GB/s) through faster memory tech (HBM3) or wider channel.